

PART B

Regional problems

Sports concussion

with PAUL McCrORY, MICHAEL MAKDISSI, GAVIN DAVIS and MICHAEL TURNER

*I've been knocked out a few times in there, but I have no idea
how many concussions I've ever had.*

Hulk Hogan

Although head injuries are common in all contact sports, the vast majority are minor. Sports in which minor head injuries are commonly observed include equestrian sports, various codes of football, boxing, gymnastics and martial arts. The incidence ranges from 0.25–9 per 1000 player hours of exposure in professional team sports. Amateur point-to-point (cross-country) jockeys have the highest concussion rate of any sport (95 concussions per 1000 player hours of exposure), followed by professional jumps and flat jockeys. With increasing media awareness, more concussions are being reported; however, direct epidemiological comparisons between sports are limited by methodological issues.

Head injuries of all levels are a medical emergency because they can prove fatal if misdiagnosed or incorrectly managed. The clinician's role in the management of acute head injuries is to (i) recognise the problem, (ii) ensure immediate resuscitation (if required) and (iii) transfer the injured athlete to the appropriate facility.

In this chapter we cover the following topics:

- definition
- prevention of concussion
- clinically relevant pathophysiology
- management of the concussed athlete
- complications of concussion
- return to play issues.

The chapter helps the reader to learn to diagnose and manage concussions in sport using the Sport Concussion Assessment Tool 3 (SCAT3™)¹ (Fig. 20.2), the Pocket Concussion Recognition Tool™ (Pocket CRT)² (Fig. 20.3) and Child SCAT3™ (Fig. 20.6).

We emphasise that the SCAT3 tools and the Pocket CRT are critical aspects of concussion management, as they are the result of more than 10 years of global collaboration and development by experts, at the 4th International Consensus Conference on Concussion in Sport held in Zurich in November 2012 (Fig. 20.1). These tools are designed to standardise the medical assessment and create a framework for the diagnosis of an acute concussion.

PRACTICE PEARL

The SCAT3, (Fig. 20.2) Child SCAT3 (Fig. 20.6) and Pocket CRT (Fig. 20.3) are derived from international conferences on concussion in sport and have been used internationally to assist clinicians and athletes reach 'quality decisions' about return to play (RTP) following concussion.

DEFINITION

'Concussion' is the term commonly used to describe a subtype of head injury. It is worth noting that the terms concussion and 'mild traumatic brain injury' (MTBI) refer to entirely different injury constructs and the terms cannot be used interchangeably. While concussion is a subset of mild traumatic brain injury, the converse is not true. A term often used in Europe is *commotio cerebri*, which has the same meaning as *concussion* in this setting.

Some publications on concussion refer to the Glasgow Coma Scale (GCS) as a means of classifying traumatic brain



Figure 20.1 The Concussion in Sport Group met in Berlin in October 2016. Their updated consensus statement and systematic reviews will be available in peer-reviewed journals including the *British Journal of Sports Medicine* (BJSM) by May 2017

COVER REPRODUCED WITH PERMISSION FROM BMJ GROUP AND CONCUSSION IN SPORT GROUP

injury (TBI). This is a validated and widely used measure of conscious state used in the assessment of TBI. Its primary role is in the monitoring of serial change in neurological status post-injury, which is critical in determining the need for further intervention. It also has a secondary role in determining the prognosis of TBI by subdividing the spectrum of injury into mild, moderate and severe TBI based on an assessment at 6 hours post injury. The scale, however, does not encompass sport concussion due to the mild nature of the symptoms involved and the fact that many concussion symptoms affecting the conscious state are well and truly resolved by 6 hours.

The 4th International Consensus Conference on Concussion in Sport re-affirmed the clinical definition of sports concussion reached previously by experts at earlier meetings (see box below).³⁻⁶

PREVENTION OF CONCUSSION

There is no good clinical evidence that currently available protective equipment will prevent concussion, although mouth guards have a clear role in preventing dental and oro-facial injury.⁷⁻¹⁰ Biomechanical studies have shown a small reduction in impact forces to the brain with the use of commercially available helmets, but these findings have not been translated to show a reduction in concussion incidence in published RCTs. For skiing and snowboarding there are a number of studies to suggest that helmets provide protection against head and facial injury and hence should be recommended for participants in alpine sports. In specific sports such as cycling, motor and equestrian sports, protective helmets may prevent other forms of head injury (e.g. skull fracture) that are related to falling on hard road surfaces and these may be an important injury prevention issue for those sports.⁷

Definition of concussion

Concussion is defined as a complex pathophysiological process affecting the brain, induced by traumatic biomechanical forces. Several common features that incorporate clinical, pathologic and biomechanical injury constructs that may be utilised in defining the nature of a concussive head injury include:

1. Concussion may be caused either by a direct blow to the head, face, neck, or elsewhere on the body with an 'impulsive' force transmitted to the head.
2. Concussion typically results in the rapid onset of short-lived impairment of neurologic function that resolves spontaneously. However, in some cases symptoms and signs may evolve over a number of minutes to hours.
3. Concussion may result in neuropathological changes, but the acute clinical symptoms largely reflect a functional disturbance rather than a structural injury and, as such, no abnormality is seen on standard structural neuroimaging studies.
4. Concussion results in a graded set of clinical symptoms that may or may not involve loss of consciousness. Resolution of the clinical and cognitive symptoms typically follows a sequential course. However, it is important to note that in some cases symptoms may be prolonged.

It is expected that future evolution of the definition of concussion will encompass clinical features, imaging changes, biomarkers, proteomics, epigenetic changes and genotype in a more comprehensive matrix of post-injury changes.

The major concern with the recommendation for helmet use in sport is the phenomenon known as 'risk compensation', whereby helmeted athletes change their playing behaviour in the misguided belief that the protective equipment will stop all injury. This is where the use of protective equipment results in behavioural change such as the adoption of more dangerous playing techniques, which can result in a paradoxical increase in injury rates. This may be a particular concern in child and adolescent athletes where head injury rates are often higher than in adult athletes.¹¹

Consideration of rule changes to reduce the head injury incidence or severity may be appropriate where a definite mechanism is implicated in a particular sport. An example of this is in soccer where research studies demonstrated that upper limb to head contact in heading contests accounted for approximately 50% of concussions.¹² Rule changes have recently been introduced in certain sports (e.g. rugby) to allow an effective off-field medical assessment to occur without compromising the athlete's welfare, affecting the flow of the game or unduly penalising the player's team. It is important to note that rule enforcement is a critical aspect of modifying injury risk in these settings and referees play an important role in this regard.

The competitive/aggressive nature of sport, which makes it fun to play and watch, should not be discouraged. However, sporting organisations should be encouraged to address violent conduct that may increase concussion risk. Fair play and respect should be supported as key elements of sport.¹³

THE INITIAL IMPACT: APPLIED PATHOPHYSIOLOGY

In a concussion, the athlete has sustained a significant impact to the brain. Although the pathophysiology of concussion remains poorly understood, the current consensus is that it reflects a disturbance of brain function rather than a structural injury. The impact sets in train a number of physiological mechanisms including a neurochemical metabolic cascade, lipid peroxidation, tau phosphorylation and amyloid folding, in addition to membrane gene-expression changes. Unlike more severe brain injury, these changes are not related to structural or neuropathological damage.

The concussed athlete, although conscious and without obvious focal neurological signs, may have impaired higher cortical function, for example, impaired short-term memory and speed of information processes. These subtle cognitive changes may only be detected by neuropsychological testing (discussed later in this chapter). While there are no gross structural changes on traditional imaging, newer imaging techniques (such as functional MRI or

connectomics) demonstrate transient alterations in brain function following concussion. The clinical significance of these results remains unclear. Subtle functional deficits are also suggested by recent work on eye movement abnormalities in the setting of TBI. These have been used as markers in other neurological diseases (e.g. Huntington's disease) and can be measured relatively easily.

Following a blow to the head, the athlete's conscious state may be altered. This may vary from simply being stunned to a significant loss of consciousness. Memory is typically affected in a concussive episode. A period of retrograde amnesia, that is, loss of memory of events prior to the incident, or anterograde amnesia, that is, loss of memory of events after the incident, may follow minor head injury.

The ability to think clearly, concentrate on tasks and process information may also be affected. Concussive symptoms such as headache, dizziness, blurred vision and nausea, as well as reduced reaction times, may place the athlete at risk of injury if they return to sport before symptoms have resolved.

Frequently, in episodes of mild concussion ('bell ringers'), the athlete will be dazed or stunned for a period of seconds only and continue playing. The other players and coaches may be unaware that a concussive episode has occurred. Any visual clues or signs and symptoms, as highlighted on the Pocket CRT, can be indicative of concussive injury. As such, it is recommended that alert medical and training staff should closely observe the actions of a player who has received a knock to the head for signs or symptoms of concussion.

It is a key aspect of concussion management that athletes are educated regarding the significance of concussion injuries and the importance of reporting symptoms to medical staff. While athletes may try to minimise problems to avoid removal from play, they may increase their risk of short and long-term problems related to the concussion.

MANAGEMENT OF THE CONCUSSED ATHLETE

Patient management is discussed in the on-field setting and with respect to return to play.

On-field management

Every athlete who sustains a head injury is at risk of having a structural brain or neck injury (e.g. brain contusion or cervical spine fracture). One of the critical roles of the initial medical assessment is to examine the player neurologically for such injuries. Having said that, it is not possible on the sidelines to absolutely exclude structural brain injury especially when a player may have ongoing cognitive impairment that limits examination or

where structural injury (e.g. subdural haematomas) may take time to develop post injury.

Clinical features that may raise concerns of structural injury include:

- the mechanism of injury, particularly if there is a high velocity of impact or collision with an unyielding body part (e.g. head to knee impact) or where the injury involves a vertical fall (e.g. fall from a horse, spear tackle in rugby)
- immediate and/or prolonged loss of consciousness
- evidence of a skull fracture and/or bleeding or cerebrospinal fluid (CSF) leak from ears/nose
- examination finding of a focal neurological deficit
- seizure
- progression of any clinical features over time.

Any deterioration in clinical state, in particular worsening headache, nausea or vomiting, or deterioration in conscious state, should raise suspicion of a structural head injury and warrant urgent investigation. Similarly, structural head injury should be kept in mind in any case where symptoms persist beyond 10 days.

A player with a suspected structural head injury requires immediate transport to a hospital with a neurosurgical unit and urgent computerised tomography (CT) brain scan to exclude intracranial pathology (e.g. haemorrhage, swelling) and to exclude/manage any associated cervical spine injury.

It is essential that all team physicians who have an on-field injury management role in their sport have formal training and certification in both first aid and trauma management. Depending upon the country concerned, there may be regional differences in certification and accreditation courses; some of the best known include Advanced Trauma Life Support (ATLS); Emergency Management of Severe Trauma (EMST); Pre-Hospital Emergency Care Course (PHECC); Pre-Hospital Trauma Life Support (PHTLS) and the British Association of Immediate Care Course (BASICS). This list is not exhaustive; all these courses deliver the skill-set required to appropriately and safely manage acute injuries. A medical degree alone is insufficient training in this regard.

The initial priorities when confronted by an acutely injured athlete are the basic principles of first aid. The simple mnemonic DRABC may be a useful aide-memoire.

- | | | |
|----------|---------------|--|
| D | Danger | Ensure that there are no immediate environmental dangers that may potentially injure the patient or treatment team. This may involve stopping play in a football match or marshalling cars on a motor racetrack. |
|----------|---------------|--|

- | | | |
|----------|--------------------|--|
| R | Response | Is the patient conscious? Can he/she talk? |
| A | Airway | Ensure a clear and unobstructed airway. Remove any mouth guard or dental device that may be present. |
| B | Breathing | Ensure the patient is breathing adequately. |
| C | Circulation | Ensure an adequate circulation. |

Once the basic first aid aspects of care have been achieved and the patient is stabilised, consideration of removal of the patient from the field to an appropriate facility for further assessment is necessary. At this time, careful assessment for the presence of a cervical spinal cord or other injury is necessary. If an alert patient complains of neck pain, has evidence of neck tenderness or deformity, or has neurological signs suggestive of a spinal injury, then neck bracing and transport on a suitable spinal frame is required. If the patient is unconscious, then a cervical injury should be assumed until proven otherwise. Airway protection takes precedence over any potential spinal injury. In this situation, the removal of helmets or other head protectors should only be performed by individuals trained in this aspect of trauma management.

The clinical management of a concussed athlete may involve the treatment of a disorientated, confused, unconscious, uncooperative or convulsing patient (see Table 20.1 for other symptoms and signs of concussion). The immediate treatment priorities remain the basic first aid principles outlined above. Once this has been established and the patient stabilised, a full medical and neurological assessment exam should follow. On site physicians are in an ideal position to initiate the critical early steps in an athlete's care to ensure optimal recovery from a head injury. Typically, with a medical doctor present on the sidelines the more detailed assessment of the injured athlete would occur in the medical rooms where a quiet and unhurried environment exists. This would be the normal situation in the case of an athlete that is conscious and without a spinal cord injury. In some situations, where no doctor is present, safe removal to a hospital emergency department is the correct approach.

When examining a concussed athlete, a full neurological examination is important. Because the major management priorities at this stage are to establish an accurate diagnosis and exclude a potentially catastrophic intracranial injury, this part of the examination should be particularly thorough. Having determined the presence of a concussive injury, the patient needs to be serially monitored until full recovery ensues. In the acute situation following a traumatic brain injury of any severity,

Table 20.1 Symptoms and signs of concussion

Symptoms	Signs
• Headache • Dizziness • Nausea or vomiting • Unsteadiness or loss of balance • Confusion • Unaware of period, opposition, score of game • Feeling 'dinged', stunned or 'dazed' • Seeing stars or flashing lights • Ringing in the ears • Double vision or blurred vision	• Loss of consciousness or impaired conscious state • Poor coordination or balance • Concussive convulsion or impact seizure • Unsteady gait or loss of balance • Slow to answer questions or follow directions • Easily distracted, poor concentration • Displaying unusual or inappropriate emotions • Vacant stare or glassy eyed • Slurred speech • Personality changes • Inappropriate playing behaviour • Significantly decreased playing ability • Double vision or blurred vision

conscious state is the key element to assess initially. This is measured in the GCS which is included in the SCAT3.

Once severe brain or cervical spine injury has been excluded, the most important components of concussion management include:

- confirming the diagnosis (includes differentiating from other pathologies, e.g. post-traumatic headaches)
- determining when the player has fully recovered so that he/she can safely return to competition.

Confirming the diagnosis

Common symptoms of concussion include headache, nausea, dizziness and balance problems, blurred vision or other visual disturbance, confusion, memory loss and a feeling of slowness or fatigue (Table 20.1). Many of these symptoms are not specific to concussion. Nevertheless, concussion should be suspected in any player who presents with any of these symptoms following a collision or direct trauma to the head. When performing an on-field assessment it is suggested that the CRT (Fig. 20.3) be utilised as a rapid means of making such assessment.

Clinical features that are more likely to suggest a diagnosis of concussion include loss of consciousness (LOC), concussive convulsions, confusion or attention deficit, memory disturbance and balance disturbance.

PRACTICE PEARL

If a concussion is suspected, the player should be removed from the playing environment, with appropriate care of the cervical spine and evaluated in a place free from distraction (e.g. medical room).

These may not be present in all cases and in some cases may present in a delayed fashion. The assessment as per the SCAT3 (Fig 20.2) form should be used and focus on:

- symptom checklist
- tests of cognitive function (e.g. Maddocks' questions, Standardised Assessment of Concussion [SAC])
- tests of balance function
- neurological examination.

One of the critical aspects of the SCAT3 is that it should be used after 10 minutes have passed following the injury. In that time, the clinician should ensure that the first aid priorities have been completed, a GCS recorded, spinal and other injuries excluded and the opportunity to review any video of the incident (if available) completed. This delay in the concussion assessment is important as, in many cases, the concussion injury is an evolving one and many athletes only begin to exhibit symptoms and signs several minutes after injury. If the SCAT3 is done too early following the injury, then the critical concussion features may be missed.

Any player with symptoms and/or evidence of a disturbance of cognitive function (e.g. LOC, balance disturbance, disorientation or cognitive deficit) can be considered to have a concussive injury. Once concussion is medically diagnosed then the player should be removed from the game or training and not return to play on that day (Fig 20.4). As a general rule, even in cases where the diagnosis is not entirely clear, a conservative approach should be adopted (*'If in doubt sit them out'*).

For practical purposes, the CRT (Fig. 20.3) can be utilised on-field or on the sideline to screen for concussion and once concussion is diagnosed, the player removed to the medical room and the full SCAT3



Name

Date/Time of Injury
Date of Assessment

Examiner

What is the SCAT3?

The SCAT3 is a standardized tool for evaluating injured athletes for concussion and can be used in athletes aged from 13 years and older. It supersedes the original SCAT and the SCAT2 published in 2005 and 2009, respectively. For younger persons, ages 12 and under, please use the Child SCAT3. The SCAT3 is designed for use by medical professionals. If you are not qualified, please use the Sport Concussion Recognition Tool. Preseason baseline testing with the SCAT3 can be helpful for interpreting post-injury test scores.

Specific instructions for use of the SCAT3 are provided on page 3. If you are not familiar with the SCAT3, please read through these instructions carefully. This tool may be freely copied in its current form for distribution to individuals, teams, groups and organizations. Any revision or any reproduction in a digital form requires approval by the Concussion in Sport Group.

NOTE: The diagnosis of a concussion is a clinical judgment, ideally made by a medical professional. The SCAT3 should not be used solely to make, or exclude, the diagnosis of concussion in the absence of clinical judgement. An athlete may have a concussion even if their SCAT3 is "normal".

What is a concussion?

A concussion is a disturbance in brain function caused by a direct or indirect force to the head. It results in a variety of non-specific signs and/or symptoms (some examples listed below) and most often does not involve loss of consciousness. Concussion should be suspected in the presence of **any one or more** of the following:

- Symptoms (e.g., headache), or
- Physical signs (e.g., unsteadiness), or
- Impaired brain function (e.g. confusion) or
- Abnormal behaviour (e.g., change in personality).

SIDELINE ASSESSMENT

Indications for Emergency Management

NOTE: A hit to the head can sometimes be associated with a more serious brain injury. Any of the following warrants consideration of activating emergency procedures and urgent transportation to the nearest hospital:

- Glasgow Coma score less than 15
- Deteriorating mental status
- Potential spinal injury
- Progressive, worsening symptoms or new neurologic signs

Potential signs of concussion?

If any of the following signs are observed after a direct or indirect blow to the head, the athlete should stop participation, be evaluated by a medical professional and **should not be permitted to return to sport the same day** if a concussion is suspected.

Any loss of consciousness?	Y	N
"If so, how long?"		
Balance or motor incoordination (stumbles, slow/laboured movements, etc)?	Y	N
Disorientation or confusion (inability to respond appropriately to questions)?	Y	N
Loss of memory	Y	N
"If so, how long?"		
"Before or after the injury?"		
Blank or vacant look:	Y	N
Visible facial injury in combination with any of the above:	Y	N

1 Glasgow coma scale (GCS)

Best eye response (E)

No eye opening	1
Eye opening in response to pain	2
Eye opening to speech	3
Eyes opening spontaneously	4

Best verbal response (V)

No verbal response	1
Incomprehensible sounds	2
Inappropriate words	3
Confused	4
Oriented	5

Best motor response (M)

No motor response	1
Extension to pain	2
Abnormal flexion to pain	3
Flexion/Withdrawal to pain	4
Localizes to pain	5
Obeys commands	6

Glasgow Coma score (E + V + M) of 15

GCS should be recorded for all athletes in case of subsequent deterioration.

2 Maddocks Score³

"I am going to ask you a few questions, please listen carefully and give your best effort."

Modified Maddocks questions (1 point for each correct answer)

What venue are we at today?	0	1
Which half is it now?	0	1
Who scored last in this match?	0	1
What team did you play last week/game?	0	1
Did your team win the last game?	0	1
Maddocks score		of 5

Maddocks score is validated for sideline diagnosis of concussion only and is not used for serial testing.

Notes: Mechanism of Injury ("tell me what happened?").

Any athlete with a suspected concussion should be REMOVED FROM PLAY, medically assessed, monitored for deterioration (i.e., should not be left alone) and should not drive a motor vehicle until cleared to do so by a medical professional. No athlete diagnosed with concussion should be returned to sports participation on the day of injury.

(a)

Figure 20.2 SCAT3™

REPRODUCED WITH PERMISSION FROM BRITISH JOURNAL OF SPORTS MEDICINE

BACKGROUND

Name: _____ Date: _____

Examiner: _____

Sport/team/school: _____ Date/time of injury: _____

Age: _____ Gender: ☐ M ☐ F

Years of education completed: _____

Dominant hand: ☐ right ☐ left ☐ neither

How many concussions do you think you have had in the past? _____

When was the most recent concussion? _____

How long was your recovery from the most recent concussion? _____

Have you ever been hospitalized or had medical imaging done for a head injury? ☐ Y ☐ N

Have you ever been diagnosed with headaches or migraines? ☐ Y ☐ N

Do you have a learning disability, dyslexia, ADD/ADHD? ☐ Y ☐ N

Have you ever been diagnosed with depression, anxiety or other psychiatric disorder? ☐ Y ☐ N

Has anyone in your family ever been diagnosed with any of these problems? ☐ Y ☐ N

Are you on any medications? If yes, please list: ☐ Y ☐ N

SCAT3 to be done in resting state. Best done 10 or more minutes post exercise.

SYMPTOM EVALUATION

3

How do you feel?

"You should score yourself on the following symptoms, based on how you feel now".

	none	mild		moderate		severe	
Headache	0	1	2	3	4	5	6
"Pressure in head"	0	1	2	3	4	5	6
Neck Pain	0	1	2	3	4	5	6
Nausea or vomiting	0	1	2	3	4	5	6
Dizziness	0	1	2	3	4	5	6
Blurred vision	0	1	2	3	4	5	6
Balance problems	0	1	2	3	4	5	6
Sensitivity to light	0	1	2	3	4	5	6
Sensitivity to noise	0	1	2	3	4	5	6
Feeling slowed down	0	1	2	3	4	5	6
Feeling like "in a fog"	0	1	2	3	4	5	6
"Don't feel right"	0	1	2	3	4	5	6
Difficulty concentrating	0	1	2	3	4	5	6
Difficulty remembering	0	1	2	3	4	5	6
Fatigue or low energy	0	1	2	3	4	5	6
Confusion	0	1	2	3	4	5	6
Drowsiness	0	1	2	3	4	5	6
Trouble falling asleep	0	1	2	3	4	5	6
More emotional	0	1	2	3	4	5	6
Irritability	0	1	2	3	4	5	6
Sadness	0	1	2	3	4	5	6
Nervous or Anxious	0	1	2	3	4	5	6

Total number of symptoms (Maximum possible 32) _____

Symptom severity score (Maximum possible 132) _____

Do the symptoms get worse with physical activity? ☐ Y ☐ N

Do the symptoms get worse with mental activity? ☐ Y ☐ N

☐ self rated ☐ self rated and clinician monitored
☐ clinician interview ☐ self rated with parent input

Overall rating: If you know the athlete well prior to the injury, how different is the athlete acting compared to his/her usual self?

Please circle one response:

☐ no different ☐ very different ☐ unsure ☐ N/A

Scoring on the SCAT3 should not be used as a stand alone method to diagnose concussion measure recovery or make decisions about an athlete's readiness to return to competition after concussion. Since signs and symptoms may evolve over time it is important to consider repeat evaluation in the acute assessment of concussion.

(b)

Figure 20.2 (cont.) SCAT3™

COGNITIVE & PHYSICAL EVALUATION

4

Cognitive assessment

Standardized Assessment of Concussion (SAC)⁴

Orientation (1 point for each correct answer)

What month is it?	0	1
What is the date today?	0	1
What is the day of the week?	0	1
What year is it?	0	1
What time is it right now? (within 1 hour)	0	1
Orientation score	of 5	

Immediate memory

List	Trial 1		Trial 2		Trial 3		Alternative word list		
elbow	0	1	0	1	0	1	candle	baby	finger
apple	0	1	0	1	0	1	paper	monkey	penny
carpet	0	1	0	1	0	1	sugar	perfume	blanket
saddle	0	1	0	1	0	1	sandwich	sunset	lemon
bubble	0	1	0	1	0	1	wagon	iron	insect
Total									

Immediate memory score total _____ of 15

Concentration: Digits Backward

List	Trial 1		Alternative digit list		
4-9-3	0	1	6-2-9	5-2-6	4-1-5
3-8-1-4	0	1	3-2-7-9	1-7-9-5	4-9-6-8
6-2-9-7-1	0	1	1-5-2-8-6	3-8-5-2-7	6-1-8-4-3
7-1-8-4-6-2	0	1	5-3-9-1-4-8	8-3-1-9-6-4	7-2-4-8-5-6
Total of 4					

Concentration: Month in Reverse Order (1 pt. for entire sequence correct)

Dec-Nov-Oct-Sept-Aug-Jul-Jun-May-Apr-Mar-Feb-Jan ☐ 0 ☐ 1

Concentration score _____ of 5

5

Neck Examination:

Range of motion _____ Tenderness _____ Upper and lower limb sensation _____ strength _____

Balance examination

Do one or both of the following tests.

Footwear (shoes, barefoot, braces, tape, etc.) _____

Modified Balance Error Scoring System (BESS) testing⁵

Which foot was tested (i.e. which is the non-dominant foot) ☐ Left ☐ Right

Testing surface (hard floor, field, etc.) _____

Condition

Double leg stance: _____ Errors _____

Single leg stance (non-dominant foot): _____ Errors _____

Tandem stance (non-dominant foot at back): _____ Errors _____

And/Or

Tandem gait

Time (best of _____ trials): _____ seconds

Coordination examination

Upper limb coordination

Which arm was tested: ☐ Left ☐ Right

Coordination score _____ of 1

8

SAC Delayed Recall⁴

Delayed recall score _____

INSTRUCTIONS

Words in *Italics* throughout the SCAT3 are the instructions given to the athlete by the tester.

Symptom Scale

"You should score yourself on the following symptoms, based on how you feel now."

To be completed by the athlete. In situations where the symptom scale is being completed after exercise, it should still be done in a resting state, at least 10 minutes post exercise.

For total number of symptoms, maximum possible is 22.

For Symptom severity score, add all scores in table, maximum possible is $22 \times 6 = 132$

SAC⁴

Immediate Memory

"I am going to test your memory. I will read you a list of words and when I am done, repeat back as many words as you can remember, in any order."

Trials 2 & 3:

"I am going to repeat the same list again. Repeat back as many words as you can remember in any order, even if you said the word before."

Complete all 3 trials regardless of score on trial 1 & 2. Read the words at a rate of one per second. **Score 1 pt. for each correct response.** Total score equals sum across all 3 trials. Do not inform the athlete that delayed recall will be tested.

Concentration Digits backward

"I am going to read you a string of numbers and when I am done, you repeat them back to me backwards, in reverse order of how I read them to you. For example, if I say 7-1-9, you would say 9-1-7."

If correct, go to next string length. If incorrect, read trial 2. **One point possible for each string length.** Stop after incorrect on both trials. The digits should be read at the rate of one per second.

Months in reverse order

"Now tell me the months of the year in reverse order. Start with the last month and go backward. So you'll say December, November.... Go ahead"

1 pt. for entire sequence correct

Delayed Recall

The delayed recall should be performed after completion of the Balance and Coordination Examination

"Do you remember that list of words I read a few times earlier? Tell me as many words from the list as you can remember in any order."

Score 1 pt. for each correct response

Balance Examination

Modified Balance Error Scoring System (BESS) testing⁵

This balance testing is based on a modified version of the Balance Error Scoring System (BESS)⁵. A stopwatch or watch with a second hand is required for this testing.

"I am now going to test your balance. Please take your shoes off, roll up your pant legs above ankle (if applicable), and remove any ankle taping (if applicable). This test will consist of three twenty second tests with different stances"

(a) Double leg stance:

"The first stance is standing with your feet together with your hands on your hips and with your eyes closed. You should try to maintain stability in that position for 20 seconds. I will be counting the number of times you move out of this position. I will start timing when you are set and have closed your eyes."

(b) Single leg stance:

"If you were to kick a ball, which foot would you use? [This will be the dominant foot] Now stand on your non-dominant foot. The dominant leg should be held in approximately 30 degrees of hip flexion and 45 degrees of knee flexion. Again, you should try to maintain stability for 20 seconds with your hands on your hips and your eyes closed. I will be counting the number of times you move out of this position. If you stumble out of this position, open your eyes and return to the start position and continue balancing. I will start timing when you are set and have closed your eyes."

(c) Tandem stance:

"Now stand heel-to-toe with your non-dominant foot in back. Your weight should be evenly distributed across both feet. Again, you should try to maintain stability for 20 seconds with your hands on your hips and your eyes closed. I will be counting the number of times you move out of this position. If you stumble out of this position, open your eyes and return to the start position and continue balancing. I will start timing when you are set and have closed your eyes."

(c)

Balance testing – types of errors

1. Hands lifted off iliac crest
2. Opening eyes
3. Step, stumble, or fall
4. Moving hip into > 30 degrees abduction
5. Lifting forefoot or heel
6. Remaining out of test position > 5 sec

Each of the 20-second trials is scored by counting the errors, or deviations from the proper stance, accumulated by the athlete. The examiner will begin counting errors only after the individual has assumed the proper start position. **The modified BESS is calculated by adding one error point for each error during the three 20-second tests. The maximum total number of errors for any single condition is 10.** If a athlete commits multiple errors simultaneously, only one error is recorded but the athlete should quickly return to the testing position, and counting should resume once subject is set. Subjects that are unable to maintain the testing procedure for a minimum of **five seconds** at the start are assigned the highest possible score, ten, for that testing condition

OPTION: For further assessment, the same 3 stances can be performed on a surface of medium density foam (e.g., approximately 50 cm x 40 cm x 6 cm)

Tandem Gait^{6,7}

Participants are instructed to stand with their feet together behind a starting line (the test is best done with footwear removed). Then, they walk in a forward direction as quickly and as accurately as possible along a 38mm wide (sports tape), 3 meter line with an alternate foot heel-to-toe gait ensuring that they approximate their heel and toe on each step. Once they cross the end of the 3m line, they turn 180 degrees and return to the starting point using the same gait. A total of 4 trials are done and the best time is retained. Athletes should complete the test in 14 seconds. Athletes fail the test if they step off the line, have a separation between their heel and toe, or if they touch or grab the examiner or an object. In this case, the time is not recorded and the trial repeated, if appropriate.

Coordination Examination

Upper limb coordination

Finger-to-nose (FTN) task:

"I am going to test your coordination now. Please sit comfortably on the chair with your eyes open and your arm (either right or left) outstretched (shoulder flexed to 90 degrees and elbow and fingers extended), pointing in front of you. When I give a start signal, I would like you to perform five successive finger to nose repetitions using your index finger to touch the tip of the nose, and then return to the starting position, as quickly and as accurately as possible."

Scoring: 5 correct repetitions in < 4 seconds = 1

Note for testers: Athletes fail the test if they do not touch their nose, do not fully extend their elbow or do not perform five repetitions. Failure should be scored as 0.

References & Footnotes

1. This tool has been developed by a group of international experts at the 4th International Consensus meeting on Concussion in Sport held in Zurich, Switzerland in November 2012. The full details of the conference outcomes and the authors of the tool are published in The BJSM Injury Prevention and Health Protection, 2013, Volume 47, Issue 5. The outcome paper will also be simultaneously co-published in other leading biomedical journals with the copyright held by the Concussion in Sport Group, to allow unrestricted distribution, providing no alterations are made.
2. McCrory P et al., Consensus Statement on Concussion in Sport – the 3rd International Conference on Concussion in Sport held in Zurich, November 2008. British Journal of Sports Medicine 2009; 43: i76-89.
3. Maddocks, DL, Dicker, GD, Saling, MM. The assessment of orientation following concussion in athletes. Clinical Journal of Sport Medicine. 1995; 5(1): 32–3.
4. McCrea M. Standardized mental status testing of acute concussion. Clinical Journal of Sport Medicine. 2001; 11: 176–181.
5. Guskiewicz KM. Assessment of postural stability following sport-related concussion. Current Sports Medicine Reports. 2003; 2: 24–30.
6. Schneiders, A.G., Sullivan, S.J., Gray, A., Hammond-Tooke, G. & McCrory, P. Normative values for 16-37 year old subjects for three clinical measures of motor performance used in the assessment of sports concussions. Journal of Science and Medicine in Sport. 2010; 13(2): 196–201.
7. Schneiders, A.G., Sullivan, S.J., Kvarnstrom, J.K., Olsson, M., Yden, T. & Marshall, S.W. The effect of footwear and sports-surface on dynamic neurological screening in sport-related concussion. Journal of Science and Medicine in Sport. 2010; 13(4): 382–386.

Figure 20.2 (cont.) SCAT3™

ATHLETE INFORMATION

Any athlete suspected of having a concussion should be removed from play, and then seek medical evaluation.

Signs to watch for

Problems could arise over the first 24–48 hours. The athlete should not be left alone and must go to a hospital at once if they:

- Have a headache that gets worse
- Are very drowsy or can't be awakened
- Can't recognize people or places
- Have repeated vomiting
- Behave unusually or seem confused; are very irritable
- Have seizures (arms and legs jerk uncontrollably)
- Have weak or numb arms or legs
- Are unsteady on their feet; have slurred speech

Remember, it is better to be safe.

Consult your doctor after a suspected concussion.

Return to play

Athletes should not be returned to play the same day of injury.

When returning athletes to play, they should be **medically cleared and then follow a stepwise supervised program**, with stages of progression

For example:

Rehabilitation stage	Functional exercise at each stage of rehabilitation	Objective of each stage
No activity	Physical and cognitive rest	Recovery
Light aerobic exercise	Walking, swimming or stationary cycling, keeping intensity 70% maximum predicted heart rate. No resistance training	Increase heart rate
Sport-specific exercise	Skating drills in ice hockey, running drills in soccer. No head-impact activities	Add movement
Non-contact training drills	Progression to more complex training drills, eg passing drills in football and ice hockey. May start progressive resistance training	Exercise coordination, and cognitive load
Full contact practice	Following medical clearance participate in normal training activities	Restore confidence and assess functional skills by coaching staff
Return to play	Normal game play	

There should be at least 24 hours (or longer) for each stage and if symptoms recur the athlete should rest until they resolve once again and then resume the program at the previous asymptomatic stage. Resistance training should only be added in the later stages

If the athlete is symptomatic for more than 10 days, then consultation by a medical practitioner who is expert in the management of concussion, is recommended.

Medical clearance should be given before return to play.

Scoring Summary:

Test Domain	Score		
	Date: _____	Date: _____	Date: _____
Number of Symptoms of 22			
Symptom Severity Score of 132			
Orientation of 5			
Immediate Memory of 15			
Concentration of 5			
Delayed Recall of 5			
SAC Total			
BESS (total errors)			
Tandem Gait (seconds)			
Coordination of 1			

Notes:

CONCUSSION INJURY ADVICE

(To be given to the **person monitoring** the concussed athlete)

This patient has received an injury to the head. A careful medical examination has been carried out and no sign of any serious complications has been found. Recovery time is variable across individuals and the patient will need monitoring for a further period by a responsible adult. Your treating physician will provide guidance as to this timeframe.

If you notice any change in behaviour, vomiting, dizziness, worsening headache, double vision or excessive drowsiness, please contact your doctor or the nearest hospital emergency department immediately.

Other important points:

- Rest (physically and mentally), including training or playing sports until symptoms resolve and you are medically cleared
- No alcohol
- No prescription or non-prescription drugs without medical supervision. Specifically:
 - No sleeping tablets
 - Do not use aspirin, anti-inflammatory medication or sedating pain killers
- Do not drive until medically cleared
- Do not train or play sport until medically cleared

Clinic phone number

(d)

Patient's name

Date/time of injury

Date/time of medical review

Treating physician

Contact details or stamp

Figure 20.2 (cont.) SCAT3™

Pocket CONCUSSION RECOGNITION TOOL™

To help identify concussion in children, youth and adults



RECOGNIZE & REMOVE

Concussion should be suspected **if one or more** of the following visible clues, signs, symptoms or errors in memory questions are present.

1. Visible clues of suspected concussion

Any one or more of the following visual clues can indicate a possible concussion:

Loss of consciousness or responsiveness
Lying motionless on ground/Slow to get up
Unsteady on feet / Balance problems or falling over / Incoordination
Grabbing/Clutching of head
Dazed, blank or vacant look
Confused/Not aware of plays or events

2. Signs and symptoms of suspected concussion

Presence of any one or more of the following signs & symptoms may suggest a concussion:

- | | |
|--------------------------|----------------------------|
| - Loss of consciousness | - Headache |
| - Seizure or convulsion | - Dizziness |
| - Balance problems | - Confusion |
| - Nausea or vomiting | - Feeling slowed down |
| - Drowsiness | - "Pressure in head" |
| - More emotional | - Blurred vision |
| - Irritability | - Sensitivity to light |
| - Sadness | - Amnesia |
| - Fatigue or low energy | - Feeling like "in a fog" |
| - Nervous or anxious | - Neck Pain |
| - "Don't feel right" | - Sensitivity to noise |
| - Difficulty remembering | - Difficulty concentrating |

© 2013 Concussion in Sport Group

3. Memory function

Failure to answer any of these questions correctly may suggest a concussion.

- "What venue are we at today?"
"Which half is it now?"
"Who scored last in this game?"
"What team did you play last week / game?"
"Did your team win the last game?"

Any athlete with a suspected concussion should be IMMEDIATELY REMOVED FROM PLAY, and should not be returned to activity until they are assessed medically. Athletes with a suspected concussion should not be left alone and should not drive a motor vehicle.

It is recommended that, in all cases of suspected concussion, the player is referred to a medical professional for diagnosis and guidance as well as return to play decisions, even if the symptoms resolve.

RED FLAGS

If ANY of the following are reported then the player should be safely and immediately removed from the field. If no qualified medical professional is available, consider transporting by ambulance for urgent medical assessment:

- | | |
|--|---------------------------------|
| - Athlete complains of neck pain | - Deteriorating conscious state |
| - Increasing confusion or irritability | - Severe or increasing headache |
| - Repeated vomiting | - Unusual behaviour change |
| - Seizure or convulsion | - Double vision |
| - Weakness or tingling/burning in arms or legs | |

Remember:

- In all cases, the basic principles of first aid (danger, response, airway, breathing, circulation) should be followed.
- Do not attempt to move the player (other than required for airway support) unless trained to do so
- Do not remove helmet (if present) unless trained to do so.

from McCrory et. al, Consensus Statement on Concussion in Sport. Br J Sports Med 47 (5), 2013

© 2013 Concussion in Sport Group

Figure 20.3 Pocket Concussion Recognition Tool™ (CRT)²
REPRODUCED WITH PERMISSION FROM *BRITISH JOURNAL OF SPORTS MEDICINE*

assessment tool used by the attending physician. If the diagnosis of concussion is confirmed following the medical assessment, the player should not be returned to play on the day and serial monitoring should be commenced immediately. New portable technologies are continually being incorporated into acute sports concussion assessment; however, research is yet to establish how effective and clinically applicable these new methods are.¹⁴

In addition to post-injury assessment, it is recommended that the SCAT3 be used by medical staff for pre-season baseline testing. This may be helpful for interpreting post-concussion test scores as it provides an objective record of possible change. More importantly, it provides an educative opportunity for the physician to highlight the importance of reporting all concussive injuries.

Does amnesia associate with injury severity?

There is renewed interest in post-traumatic amnesia and its role as a surrogate measure of injury severity.

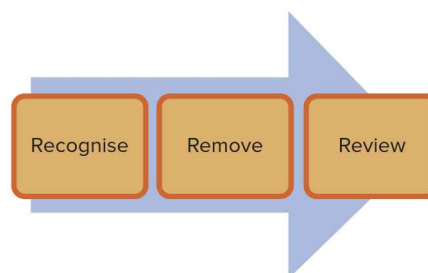


Figure 20.4 Recognition of concussion, removal from play, and review and assessment of the athlete are the key steps in management

Post-traumatic amnesia can be separated into retrograde amnesia (loss of memory prior to injury) and anterograde amnesia (loss of memory post injury). Virtually all concussed athletes will have periods of amnesia, however brief, whether they were unconscious or not.

PRACTICE PEARL

In broad terms, the duration of post-traumatic amnesia does not reflect the severity of concussion or mild TBI. This contrasts with moderate to severe traumatic brain injury where post-traumatic amnesia is a prognostic factor and should be assessed in all cases.

Note that retrograde amnesia varies depending on when it is measured after injury; thus, it is a particularly poor indicator of injury severity.

The number and duration of the clinical post-concussive symptoms may be more important than the presence or duration of amnesia alone. The overall clinical management and return to play depends on the presence and recovery of all symptoms and signs.

Does an acutely concussed athlete need to go to hospital or have urgent neuroimaging?

The treating clinician may face the decision of whether the athlete should be referred on to a hospital emergency facility or for urgent neuroimaging. In general terms, an uncomplicated concussion does not need routine neuroimaging. Imaging, however, has a role in the exclusion of suspected intracranial injury and the indications are listed in the box. Apart from 'cookbook' type approaches, referral to such a centre depends on the experience, ability and competency of the physician at hand. If the team physician happens to be an SEM physician, neurologist or neurosurgeon experienced in concussion management, the clinical referral pathways will be different to a family practitioner called to assist at a football match after an injury has occurred.

PRACTICE PEARL

The overall approach should be 'when in doubt, refer'.

Other diagnostic tests such as biomarkers, EEG and functional MRI brain scanning do not have a role in the early diagnosis and management of a concussion injury, except in an experimental or research context.¹⁵ Structural neuroimaging such as CT and MRI does not clinically add to a diagnosis of concussion; however, these have an important role in excluding more significant injuries.

How should acute concussion be graded?

There is no reliable or scientifically validated system of grading the severity of sports-related concussion. At the present time, there are at least 45 published anecdotal

Indications for urgent imaging or hospital referral

Any player who has or develops the following:

- fractured skull
- penetrating skull trauma
- deterioration in conscious state following injury
- focal neurological signs
- confusion or impairment of consciousness >30 minutes
- loss of consciousness >1 minute
- persistent vomiting or increasing headache post injury
- any convulsive movements
- more than one episode of concussive injury in a match or training session
- where there is assessment difficulty (e.g. an intoxicated patient)
- all children with head injuries
- high-risk patients (e.g. haemophilia, anticoagulant use)
- inadequate post injury supervision
- high-risk injury mechanism (e.g. high velocity impact, missile injury).

severity scales. The danger is that athletes and/or their coaches may 'shop around' for a scale that is not in their best medical interests. At the end of the day, good clinical judgment should prevail over written guidelines. At all four international conferences on concussion in sport to date, the expert committee has strongly recommended that combined measures of recovery should be used to assess injury severity and guide individual decisions on return to play.⁵ Individual assessment was strongly re-endorsed at the 2012 Zurich meeting.⁶

Determining when the player can return safely to competition

Return-to-play decisions remain difficult. Expert consensus guidelines recommend that players should **not** return to competition until they have recovered **completely** from their concussive injury. The majority (80-90%) of clinical symptoms and cognitive deficits resolve in a relatively short 7-10 day period. MRI diffusion or connectomic radiological changes may still be present for longer; however, the significance of such changes is unknown.¹⁶⁻¹⁸ Children and adolescents may take longer to recover from concussive injuries.¹⁹

Currently, there is no single gold standard measure of brain disturbance and recovery following concussion. It should be noted that most athletes wanting to return to play will report 'feeling fine'. Therefore, clinicians should incorporate both subjective and objective measures to determine readiness to return to play (RTP). In practical

terms, this can involve a multifaceted and multidisciplinary clinical approach, which includes assessment of symptoms, signs (such as balance) and cognitive function.

Return to play on the day of injury

PRACTICE PEARL

The Zurich consensus group reaffirmed the previous general management principle that no return to play on the day should be contemplated for a concussed athlete.

There is published evidence in high school and college athletes that RTP on the day of injury resulted in delayed and prolonged cognitive deterioration. It is not within the scope or expertise of a physiotherapist, trainer or other non-medical person to manage a concussive injury or determine the timing of return to play. A player should never RTP while symptomatic: 'When in doubt, sit them out!'

Return to play during the subsequent week(s)

Whether or not to allow the concussed athlete to return to training and then competition is one of the most difficult decisions the SEM physician must make. In minor cases of concussion, where all symptoms resolve quickly and there is no sign of cortical dysfunction, or evidence of impairment of short-term memory or information processing, the player may be allowed to return once recovered. Neuropsychological testing may be used to confirm full recovery. A player permitted to return to play should be closely observed for any signs of impaired function. A multifaceted clinical approach is used to manage players following injury (Fig. 20.5). In practical terms, this involves five sequential steps:

1. a brief period of cognitive and physical rest to facilitate recovery
2. monitoring for recovery of post-concussion symptoms and signs
3. the use of neuropsychological tests to estimate recovery of cognitive function
4. a graduated return to activity with monitoring for recurrence of symptoms
5. a final medical clearance before resuming full contact training and/or playing.

Period of cognitive and physical rest to facilitate recovery

Early rest is important to allow recovery following a concussive injury; however, prolonged rest may be counterproductive.^{20, 21} In the immediate post-concussion stage (24–48 hours), physical activity, physiological stress (e.g. altitude and flying) and

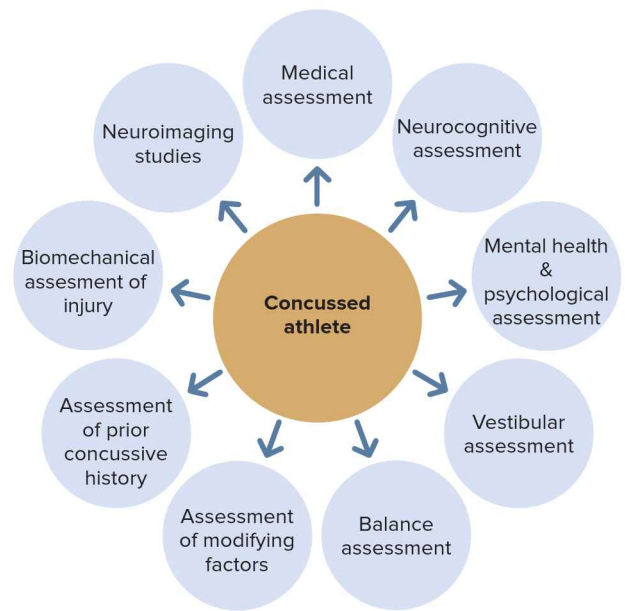


Figure 20.5 Multimodality concussion assessment

cognitive loads (e.g. studying, video games or computer work) can all worsen symptoms and possibly delay recovery. Players should be advised to rest from these activities in the early stages after a concussive injury, especially while symptomatic (see 'Concussion injury advice', p. 4 of the SCAT3). Similarly, the use of alcohol, narcotic analgesics, anti-inflammatory medication or sedatives can exacerbate symptoms following head trauma, delay recovery or mask deterioration, and should also be avoided. Specific advice should also be given on avoidance of activities that place the individual at risk of further injury (e.g. driving).

The recovery and outcome of concussion may be modified by a number of factors that may require more sophisticated management strategies. The Zurich consensus panel agreed that a range of 'modifying' factors may influence the investigation and management of concussion and, in some cases, may predict the potential for prolonged or persistent symptoms.^{6, 22} These modifiers would also be important to consider in a detailed concussion history and are outlined in Table 20.2.

When these modifying influences are present, simple RTP advice may be inappropriate. It may be wise to consider additional investigations including: formal neuropsychological testing, balance assessment and neuroimaging. It is envisioned that athletes with such modifying features would be managed in a multidisciplinary manner coordinated by a physician with specific expertise in the management of concussive injury.

Table 20.2 Factors that influence whether investigation or more sophisticated management (e.g. referral to a physician with expertise in concussion management) is indicated

Factors	Modifier
Symptoms	Number Duration (>10 days) Severity
Signs	Prolonged LOC (>1 min), amnesia
Sequelae	Concussive convulsions
Temporal	Frequency—repeated concussions over time Timing—injuries close together in time 'Recency'—recent concussion or TBI
Threshold	Repeated concussions occurring with progressively less impact force or slower recovery after each successive concussion.
Age	Child and adolescent (<18 years old)
Co- and pre-morbidities	Migraine, depression or other mental health disorders, attention deficit hyperactivity disorder (ADHD), learning disabilities (LD), sleep disorders
Medication	Psychoactive drugs, anticoagulants
Behaviour	Dangerous style of play
Sport	High-risk activity, contact and collision sport, high sporting level

Monitoring for recovery of post-concussion symptoms and signs

The symptoms of concussion are dynamic and evolve over time. It is important that players who are suspected of having a concussion be monitored over time to assess for delayed symptom onset. Monitoring of post-concussion symptoms and signs can be facilitated by the use of the SCAT3.

Use of neuropsychological tests to estimate recovery of cognitive function

Cognitive deficits associated with concussion are typically subtle and may exist in a number of domains. Common

deficits that follow concussion in sport include reduced attention and ability to process information, slowed reaction times and impaired memory.

The use of neuropsychological tests in the management of concussion overcomes the reliance on subjective symptoms, which are known to be poorly recognised and variably reported. It allows detection of cognitive deficits, which have been observed to outlast symptoms in many cases of concussion.

There are a number of levels of complexity of cognitive testing, including:

- formal neuropsychological testing
- screening computerised cognitive test batteries
- basic paper-and-pencil evaluation (i.e. SCAT3).

Formal neuropsychological testing performed by trained neuropsychologists remains the clinical best practice standard for the assessment of cognitive function.²³ Formal testing is logistically impractical for routine use following all concussive injuries, but is recommended in any case where there is uncertainty about recovery or in difficult cases (e.g. prolonged recovery).

Screening computerised cognitive tests provide a practical alternative for the assessment of cognitive recovery. Ideally, the tests should be compared to the individual's own pre-injury baseline. A number of screening computerised cognitive test batteries have been validated for use following concussion in sport and are readily available. These include test platforms such as CogState Sport/Axon Sports (www.cogstate.com/go/Sport), ImPACT (www.impacttest.com) and the US military-developed tool Automated Neuropsychological Assessment Metrics (ANAM).

Computerised tests provide a quick, valid and reliable measure of cognitive recovery following a concussive injury. Furthermore, routine use of computerised screening tests in the pre-season facilitates screening of players for cognitive deterioration over time.

Basic paper-and-pencil cognitive tests (e.g. SCAT3, Fig. 20.2) are the quickest and simplest of the cognitive screening tests; however, they are the least sensitive to the subtle cognitive changes that accompany concussion. In cases where the concussion has resulted in brief symptoms and clinically the player has recovered well, basic paper-and-pencil cognitive tests can be used to provide an estimate of cognitive function. The use of a basic paper-and-pencil evaluation should be combined with a conservative return to play approach and careful monitoring of symptoms as the player progresses through a graduated RTP program.

Overall, it is important to remember that neuropsychological testing is only one component of assessment and therefore should not be the sole basis of management decisions. Neuropsychological testing

does not replace the need for a full history and clinical/neurological assessment.

Graduated return to activity

Following a concussive injury, players should be returned to play in a graduated fashion once clinical features have resolved and cognitive function returned to 'baseline'. When considering RTP the athlete should be off any medications prescribed in the management of the concussion at the time of considering commencement of the rehabilitation phase or at the final medical assessment.

In accordance with current consensus guidelines, there is no mandatory period of time that a player must be withheld from play following a concussion. However, at minimum, a player must be symptom free at rest and with exertion and determined to have returned to baseline level of cognitive performance.

The Zurich consensus statement recommends a stepwise Graduated Return to Play Protocol (Table 20.3). If a player remains asymptomatic for 24 hours at level 1, they may progress to level 2. They are allowed to advance provided that they remain asymptomatic. Using this protocol, an athlete should take approximately 1 week before returning to normal game play. If any symptoms surface during the progression, players should drop back to the previous level in which they were asymptomatic for a further 24 hours before attempting to progress again.

A final medical clearance before resuming full contact training and/or playing

A player who has suffered from a concussive injury must not be allowed to return to play before having a medical

clearance. In every case, the decision regarding the timing of return to training should be made by a medical doctor with experience in concussive injuries. This assessment is multidimensional and based on evidence of resolution of the athlete's symptoms, physical signs and cognitive deficit. Ideally, concussed players should be examined by an experienced medical practitioner with the decision about RTP based on the clinical findings and, if possible, neuropsychological testing. In general, a more conservative approach (i.e. longer time to return to sport) is used in cases where there is any uncertainty about the player's recovery ('If in doubt, sit them out').

In a number of contact sports, such as boxing and rugby (previously), authorities have legislated for a mandatory exclusion period from competition for concussed players. While the intent of such a policy is praiseworthy, an arbitrary exclusion period is hard to justify scientifically as each episode of concussion requires individual evaluation. For some players, the period of exclusion will be too long and for other players not long enough. More importantly, the assumption that a player has recovered simply because a prescribed period of time has passed has the potential to lead to premature return to play and resultant problems.

The risk of premature return to play and concussion sequelae

There are several risks associated with premature RTP.

Risk of further injury

The principal concern of premature RTP of a concussed athlete is that due to the impaired cognitive function (e.g. slowed information processing, reduced attention) the

Table 20.3 Graduated Return to Play Protocol

Rehabilitation stage	Functional exercise at each stage of rehabilitation	Objective at each stage
1. No activity	Complete physical (no training, playing, exercise, weights etc.) and cognitive rest (no television, extensive reading, video games etc.)	Recovery* <i>Beware of exertion with activities of daily living and caution regarding daytime sleep</i>
2. Light aerobic exercise	Walking, swimming or stationary cycling keeping intensity <70% HR max. No resistance training	Increase heart rate
3. Sport-specific exercise	Skating drills in ice hockey, running drills in soccer. No head-impact activities	Add movement
4. Non-contact training drills	Progression to more complex training drills, e.g. passing drills in hockey and football	Exercise, coordination and cognitive load
5. Full contact practice	Following medical clearance participate in normal training activities	Restore confidence and assess functional skills by coaching staff
6. Return to play	Normal game play	

Reproduced with permission from Table 1: Zurich consensus statement in BJSM³

athlete will sustain further injury (both concussive and other) when returning to a dangerous playing environment.

Second impact syndrome

Second impact syndrome (SIS) is frequently mentioned in the concussion literature, but surprisingly, has little scientific evidence for its existence. It is a term used to describe the potential catastrophic consequences resulting from a second concussive blow to the head before an individual has fully recovered from the symptoms of a previous concussion. The second head injury is believed to result in loss of cerebrovascular auto-regulation, which in turn leads to brain swelling secondary to increased cerebral blood flow. Mortality in this condition approaches 100%. The evidence that repeated concussion was a risk factor for SIS has been critically reviewed and published cases of SIS were classified as definite, probable, possible or not SIS according to four criteria.²⁴

Seventeen published cases of SIS were identified from the literature. None were classified as 'definite' SIS, five were considered to be 'probable' SIS cases and 12 were classified as 'not' SIS primarily because there was an absence of a witnessed second impact. In addition, the veracity of teammate recall of concussive episodes, which is often the basis of a 'first impact' in such cases, was shown to be unreliable. Based on these results, the investigators concluded that there is a lack of evidence to support the claim that the second impact is a risk factor for diffuse cerebral swelling.

In the review paper,²⁴ the central issue is whether repeated concussion was a risk factor for cerebral swelling which is the putative definition of SIS. There is published evidence that acute (and delayed) brain swelling may occur following a single blow to the head, in association with a structural injury such as a subdural haematoma and in disorders of calcium channels, suggesting a possible genetic basis for some of these cases. Such events are virtually only seen in children and adolescents.

Concussive convulsions

A variety of immediate motor phenomena (e.g. tonic posturing) or convulsive movements may accompany a concussion. Although dramatic, these clinical features are generally benign and require no specific management beyond the standard treatment of the underlying concussive injury. These dramatic phenomena are a non-epileptic manifestation of concussion.²⁵

Prolongation of symptoms

If a player recommences playing while symptomatic, post-concussive symptoms may be prolonged. It is important to be aware that studies have demonstrated measurable cognitive deficits in individuals even after the

symptoms have resolved, which argues for an increasingly conservative RTP strategy.¹⁷ Premature RTP may also increase the chance of developing the 'post-concussive syndrome', in which fatigue, difficulty in concentration and headaches persist for some time, often months, following the original injury. This syndrome is uncommon in sport. These patients should undergo formal neuropsychological testing as well as an MRI brain scan and be evaluated by a physician experienced in concussion management. Other potential contributing factors should be assessed and treated where appropriate, for example vestibular system disturbance.

Chronic traumatic encephalopathy (CTE)

Recent publications have suggested that US footballers may suffer similar neuropathological risks to boxers. Pathological case reports and cross-sectional studies have suggested that retired NFL footballers, who have had recurrent concussions during their sporting careers, disproportionately suffer from mild cognitive impairment, depression and other mental health problems. What is becoming increasingly clear from a number of diverse lines of research is that a percentage of footballers seem to suffer chronic or long-term sequelae.²⁶ While there has been an association demonstrated between recurrent head trauma and appearance of abnormal Tau protein in post-mortem assessment of brains, whether this is a consequence of head injury, the effect of brain injury on ageing, or the development of neurodegenerative disease unrelated to sports participation is unclear. At this time, very little is known about what type, frequency or amount of trauma is necessary to induce the accumulation of pathological proteins in the brain and more importantly why only a small number of athletes are at risk of CTE.¹⁸ Nevertheless, until these issues are resolved, it is recommended that conservative management strategies are used to ensure player welfare.

Mental health issues

Mental health issues (e.g. depression) have been reported as a long-term consequence of traumatic brain injury including sports-related concussion. Neuroimaging studies using functional MRI (fMRI) suggest that a depressed mood following concussion may reflect an underlying pathophysiological abnormality consistent with a limbic-frontal model of depression. All players with ongoing symptoms or a prolonged clinical course should be screened for depression.

THE 'DIFFICULT' CONCUSSION

A 'difficult' concussion can be defined as one in which clinical recovery falls outside the expected window (i.e. 10–14 days).¹⁷ The incidence of persistent symptoms

following sports-related concussion varies from about 10–30% depending on the cohort being investigated and definitions used.

Clinical assessment

Common persistent symptoms include headache, dizziness, mood changes (e.g. depression/anxiety), 'difficulty concentrating', irritability, 'fatigue or low energy', 'difficulty sleeping' or 'feeling not quite right, in a fog or slowed down'. The symptoms are non-specific and overlap with other conditions such as structural head injury, cervicogenic headaches and anxiety/depression. Clinical assessment, therefore, requires careful history and a detailed, multimodal examination (including neurological, cognitive function, cervical spine, balance and vestibular function).

Role of investigations

Formal neuropsychological testing is useful to identify persistent cognitive deficits, monitor recovery and help guide return to school or work plans.

Conventional neuroimaging (CT or MRI) should be considered in any athlete with persistent symptoms where there is any suspicion of an underlying structural brain injury. Advanced neuroimaging and other investigation techniques (e.g. diffusion tensor imaging, fMRI, magnetic resonance spectroscopy, electroencephalography) have demonstrated changes in cohorts of patients with persistent symptoms. Changes have also been observed in subjects well after clinical recovery following sports-related concussion. At present, the clinical significance of these changes remains unclear. Consequently, the use of advanced investigation techniques is currently not recommended in the routine management of athletes with concussion.

Treatment

A brief period of rest is important in the acute period following sports-related concussion. There is no evidence, however, that prolonged rest is beneficial for athletes with persistent symptoms. Preliminary studies demonstrate that an active rehabilitation program ('sub-symptom threshold activity') may be useful for the management of cases where symptoms are prolonged.^{27–28} Cervicovestibular rehabilitation may also be useful to facilitate recovery in cases with persistent symptoms.²⁹

Other treatments may be used to manage specific symptoms. For example, manual or physical therapy may be used to treat myofascial pain or neck trigger points contributing to headaches; cognitive therapy may be useful for some patients with persistent cognitive

symptoms; and meditation, biofeedback or psychological therapy may benefit those who have problems with psychological symptoms.

Medications are often used to treat specific symptoms (e.g. analgesics for headache, anti-depressants), although there is limited evidence for their use in the setting of persistent post-concussive symptoms.

Overall, the difficult concussion should be managed in a multi-disciplinary manner. Ideally, this is in the setting of a concussion clinic with access to expertise in a wide range of areas.

CHILDREN AND CONCUSSION IN SPORT

The CRT (Fig. 20.3) aims to identify concussion in children, youth and adults. It is simple to use and is designed so that anyone can use it. This tool is particularly useful at children's sports events that may not have a recognised health professional present. If a concussive injury is suspected, the child should be removed from play and sent urgently for medical evaluation.¹⁶ It is never appropriate for a child or adolescent athlete with concussion to return to play on the same day as the injury regardless of the level of athletic performance.

For medical assessments in this setting, the SCAT3 (Fig. 20.2) is appropriate for use in children and adolescents older than 13 years of age and the Child SCAT3 (Fig. 20.6) for 12 years and under. An additional consideration in assessing the child or adolescent athlete with a concussion is the need to include parent input as well as teacher and school input when appropriate.

The decision to use neuropsychological testing is broadly the same as the adult assessment paradigm. However, timing of testing may differ in order to assist planning in school and home management (and may be performed while the child is still symptomatic). If cognitive testing is performed then it must be developmentally sensitive until late teen years due to the ongoing cognitive maturation that occurs during this period which, in turn, makes the utility of comparison to either the person's own baseline or to population norms limited. In this age group it is more important to consider the use of trained neuropsychologists to interpret assessment data, particularly in children with learning disorders and/or ADHD who may need more sophisticated assessment strategies.

Children should not be returned to practice or play until clinically completely symptom free, which may require a longer time frame than for adults. In addition, the concept of 'cognitive rest' is highlighted with special reference to a child's need to limit exertion with activities of daily living and to limit scholastic and other cognitive stressors (e.g. text messaging, video games) while symptomatic.

Child-SCAT3™

Sport Concussion Assessment Tool for children ages 5 to 12 years

For use by medical professionals only







What is childSCAT3?

The ChildSCAT3 is a standardized tool for evaluating injured children for concussion and can be used in children aged from 5 to 12 years. It supersedes the original SCAT and the SCAT2 published in 2005 and 2009, respectively. For older persons, ages 13 years and over, please use the SCAT3. The ChildSCAT3 is designed for use by medical professionals. If you are not qualified, please use the Sport Concussion Recognition Tool. Pre-season baseline testing with the ChildSCAT3 can be helpful for interpreting post-injury test scores.

Specific instructions for use of the ChildSCAT3 are provided on page 3. If you are not familiar with the ChildSCAT3, please read through these instructions carefully. This tool may be freely copied in its current form for distribution to individuals, teams, groups and organizations. Any revision and any reproduction in a digital form require approval by the Concussion in Sport Group.

NOTE: The diagnosis of a concussion is a clinical judgment, ideally made by a medical professional. The ChildSCAT3 should not be used solely to make, or exclude, the diagnosis of concussion in the absence of clinical judgement. An athlete may have a concussion even if their ChildSCAT3 is "normal".

What is a concussion?

A concussion is a disturbance in brain function caused by a direct or indirect force to the head. It results in a variety of non-specific signs and/or symptoms (like those listed below) and most often does not involve loss of consciousness. Concussion should be suspected in the presence of any one or more of the following:

- Symptoms (e.g., headache), or
- Physical signs (e.g., unsteadiness), or
- Impaired brain function (e.g., confusion) or
- Abnormal behaviour (e.g., change in personality).

SIDELINE ASSESSMENT

Indications for Emergency Management

NOTE: A hit to the head can sometimes be associated with a more severe brain injury. If the concussed child displays any of the following, then do not proceed with the ChildSCAT3; instead activate emergency procedures and urgent transportation to the nearest hospital.

- Glasgow Coma score less than 15
- Deteriorating mental status
- Potential spinal injury
- Progressive, worsening symptoms or new neurologic signs
- Persistent vomiting
- Evidence of skull fracture
- Post-traumatic seizures
- Coagulopathy
- History of Neurosurgery (eg Shunt)
- Multiple injuries

1 Glasgow coma scale (GCS)

Best eye response (E)	
No eye opening	1
Eye opening in response to pain	2
Eye opening to speech	3
Eyes opening spontaneously	4
Best verbal response (V)	
No verbal response	1
Incomprehensible sounds	2
Inappropriate words	3
Confused	4
Oriented	5
Best motor response (M)	
No motor response	1
Extension to pain	2
Abnormal flexion to pain	3
Flexion/Withdrawal to pain	4
Localizes to pain	5
Obeys commands	6
Glasgow Coma score (E + V + M)	of 15

GCS should be recorded for all athletes in case of subsequent deterioration.

(a)

Figure 20.6 Child SCAT3™

REPRODUCED WITH PERMISSION FROM BRITISH JOURNAL OF SPORTS MEDICINE

Potential signs of concussion?

If any of the following signs are observed after a direct or indirect blow to the head, the child should stop participation, be evaluated by a medical professional and **should not be permitted to return to sport the same day** if a concussion is suspected.

Any loss of consciousness?	Y	N
"If so, how long?"		
Balance or motor incoordination (stumbles, slow/laboured movements, etc.)?	Y	N
Disorientation or confusion (inability to respond appropriately to questions)?	Y	N
Loss of memory.	Y	N
"If so, how long?"		
"Before or after the injury?"		
Blank or vacant look:	Y	N
Visible facial injury in combination with any of the above.	Y	N

2 Sideline Assessment – child-Maddocks Score³

"I am going to ask you a few questions, please listen carefully and give your best effort."

Modified Maddocks questions (1 point for each correct answer)

Where are we at now?	0	1
Is it before or after lunch?	0	1
What did you have last lesson/class?	0	1
What is your teacher's name?	0	1
child-Maddocks score	of 4	

Child-Maddocks score is for sideline diagnosis of concussion only and is not used for serial testing.

Any child with a suspected concussion should be REMOVED FROM PLAY, medically assessed and monitored for deterioration (i.e., should not be left alone). No child diagnosed with concussion should be returned to sports participation on the day of injury.

BACKGROUND

Name:	Date/Time of Injury:
Examiner:	Date of Assessment:
Sport/team/school:	
Age:	Gender: M F
Current school year/grade:	
Dominant hand:	right left neither
Mechanism of Injury ("tell me what happened?"):	

For Parent/carer to complete:

How many concussions has the child had in the past?

When was the most recent concussion?

How long was the recovery from the most recent concussion?

Has the child ever been hospitalized or had medical imaging done (CT or MRI) for a head injury?	Y	N
Has the child ever been diagnosed with headaches or migraines?	Y	N
Does the child have a learning disability, dyslexia, ADD/ADHD, seizure disorder?	Y	N
Has the child ever been diagnosed with depression, anxiety or other psychiatric disorder?	Y	N
Has anyone in the family ever been diagnosed with any of these problems?	Y	N
Is the child on any medications? If yes, please list:	Y	N

SYMPTOM EVALUATION

3 Child report

Name: _____

	never	rarely	sometimes	often
I have trouble paying attention	0	1	2	3
I get distracted easily	0	1	2	3
I have a hard time concentrating	0	1	2	3
I have problems remembering what people tell me	0	1	2	3
I have problems following directions	0	1	2	3
I daydream too much	0	1	2	3
I get confused	0	1	2	3
I forget things	0	1	2	3
I have problems finishing things	0	1	2	3
I have trouble figuring things out	0	1	2	3
It's hard for me to learn new things	0	1	2	3
I have headaches	0	1	2	3
I feel dizzy	0	1	2	3
I feel like the room is spinning	0	1	2	3
I feel like I'm going to faint	0	1	2	3
Things are blurry when I look at them	0	1	2	3
I see double	0	1	2	3
I feel sick to my stomach	0	1	2	3
I get tired a lot	0	1	2	3
I get tired easily	0	1	2	3

Total number of symptoms (Maximum possible 20) _____

Symptom severity score (Maximum possible 20 x 3 = 60) _____

☐ self rated ☐ clinician interview ☐ self rated and clinician monitored

4 Parent report

The child

	never	rarely	sometimes	often
has trouble sustaining attention	0	1	2	3
is easily distracted	0	1	2	3
has difficulty concentrating	0	1	2	3
has problems remembering what he/she is told	0	1	2	3
has difficulty following directions	0	1	2	3
tends to daydream	0	1	2	3
gets confused	0	1	2	3
is forgetful	0	1	2	3
has difficulty completing tasks	0	1	2	3
has poor problem solving skills	0	1	2	3
has problems learning	0	1	2	3
has headaches	0	1	2	3
feels dizzy	0	1	2	3
has a feeling that the room is spinning	0	1	2	3
feels faint	0	1	2	3
has blurred vision	0	1	2	3
has double vision	0	1	2	3
experiences nausea	0	1	2	3
gets tired a lot	0	1	2	3
gets tired easily	0	1	2	3

Total number of symptoms (Maximum possible 20) _____

Symptom severity score (Maximum possible 20 x 3 = 60) _____

Do the symptoms get worse with physical activity? ☐ Y ☐ N

Do the symptoms get worse with mental activity? ☐ Y ☐ N

☐ parent self rated ☐ clinician interview ☐ parent self rated and clinician monitored

Overall rating for parent/teacher/coach/carer to answer.
How different is the child acting compared to his/her usual self?

Please circle one response:

☐ no different ☐ very different ☐ unsure ☐ N/A

Name of person completing Parent-report: _____

Relationship to child of person completing Parent-report: _____

Scoring on the ChildSCAT3 should not be used as a stand-alone method to diagnose concussion, measure recovery or make decisions about an athlete's readiness to return to competition after concussion.

(b)

Figure 20.6 (cont.) Child SCAT3™

COGNITIVE & PHYSICAL EVALUATION

5 Cognitive assessment
Standardized Assessment of Concussion – Child Version (SAC-C)⁴

Orientation (1 point for each correct answer)

What month is it?	0	1
What is the date today?	0	1
What is the day of the week?	0	1
What year is it?	0	1

Orientation score _____ of 4

Immediate memory

List	Trial 1	Trial 2	Trial 3	Alternative word list			
elbow	0	1	0	1	candle	baby	finger
apple	0	1	0	1	paper	monkey	penny
carpet	0	1	0	1	sugar	perfume	blanket
saddle	0	1	0	1	sandwich	sunset	lemon
bubble	0	1	0	1	wagon	iron	insect
Total							

Immediate memory score total _____ of 15

Concentration: Digits Backward

List	Trial 1	Alternative digit list			
6-2	0	1	5-2	4-1	4-9
4-9-3	0	1	6-2-9	5-2-6	4-1-5
3-8-1-4	0	1	3-2-7-9	1-7-9-5	4-9-6-8
6-2-9-7-1	0	1	1-5-2-8-6	3-8-5-2-7	6-1-8-4-3
7-1-8-4-6-2	0	1	5-3-9-1-4-8	8-3-1-9-6-4	7-2-4-8-5-6
Total of 5					

Concentration: Days in Reverse Order (1 pt. for entire sequence correct)

Sunday-Saturday-Friday-Thursday-Wednesday-Tuesday-Monday	0	1
--	---	---

Concentration score _____ of 6

6 Neck Examination:
Range of motion Tenderness Upper and lower limb sensation & strength
Findings: _____

7 Balance examination
Do one or both of the following tests.
Footwear (shoes, barefoot, braces, tape, etc.) _____

Modified Balance Error Scoring System (BESS) testing⁵
Which foot was tested (i.e. which is the non-dominant foot) ☐ Left ☐ Right
Testing surface (hard floor, field, etc.) _____

Condition

Double leg stance:	Errors
Tandem stance (non-dominant foot at back):	Errors

Tandem gait^{6,7}
Time taken to complete (best of 4 trials): _____ seconds
If child attempted, but unable to complete tandem gait, mark here ☐

8 Coordination examination
Upper limb coordination
Which arm was tested: ☐ Left ☐ Right
Coordination score _____ of 1

9 SAC Delayed Recall⁴
Delayed recall score _____ of 5

Since signs and symptoms may evolve over time, it is important to consider repeat evaluation in the acute assessment of concussion.

INSTRUCTIONS

Words in *italics* throughout the ChildSCAT3 are the instructions given to the child by the tester.

Sideline Assessment – child-Maddocks Score

To be completed on the sideline/in the playground, immediately following concussion. There is no requirement to repeat these questions at follow-up.

Symptom Scale⁸

In situations where the symptom scale is being completed after exercise, it should still be done in a resting state, at least 10 minutes post exercise.

On the day of injury

- the child is to complete the Child Report, according to how he/she feels now.

On all subsequent days

- the child is to complete the Child Report, according to how he/she feels today, **and**
- the parent/carer is to complete the Parent Report according to how the child has been over the previous 24 hours.

Standardized Assessment of Concussion – Child Version (SAC-C)⁴

Orientation

Ask each question on the score sheet. A correct answer for **each question scores 1 point**. If the child does not understand the question, gives an incorrect answer, or no answer, then the score for that question is 0 points.

Immediate memory

"I am going to read you a string of memory. I will read you a list of words and when I am done, repeat back as many words as you can remember, in any order."

Trials 2 & 3:

"I am going to repeat the same list again. Repeat back as many words as you can remember in any order, even if you said the word before"

Complete all 3 trials regardless of score on trial 1 & 2. Read the words at a rate of one per second. **Score 1 pt. for each correct response.** Total score equals sum across all 3 trials. Do not inform the child that delayed recall will be tested.

Concentration

Digits Backward:

"I am going to read you a string of numbers and when I am done, you repeat them back to me backwards, in reverse order of how I read them to you. For example, if I say 7-1, you would say 1-7."

If correct, go to next string length. If incorrect, read trial 2. **One point possible for each string length.** Stop after incorrect on both trials. The digits should be read at the rate of one per second.

Days in Reverse Order:

"Now tell me the days of the week in reverse order. Start with Sunday and go backward. So you'll say Sunday, Saturday ... Go ahead"

1 pt. for entire sequence correct

Delayed recall

The delayed recall should be performed after completion of the Balance and Coordination Examination.

"Do you remember that list of words I read a few times earlier? Tell me as many words from the list as you can remember in any order"

Circle each word correctly recalled. **Total score equals number of words recalled.**

Balance examination

These instructions are to be read by the person administering the childSCAT3, and each balance task **should be demonstrated to the child**. The child should then be asked to copy what the examiner demonstrated.

Modified Balance Error Scoring System (BESS) testing⁵

This balance testing is based on a modified version of the Balance Error Scoring System (BESS)⁵. A stopwatch or watch with a second hand is required for this testing.

"I am now going to test your balance. Please take your shoes off, roll up your pant legs above ankle (if applicable), and remove any ankle taping (if applicable). This test will consist of two different parts:"

(a) Double leg stance:

The first stance is standing with the feet together with hands on hips and with eyes closed. The child should try to maintain stability in that position for 20 seconds. You should inform the child that you will be counting the number of times the child moves out of this position. You should start timing when the child is set and the eyes are closed.

(b) Tandem stance:

Instruct the child to stand heel-to-toe with the non-dominant foot in the back. Weight should be evenly distributed across both feet. Again, the child should try to maintain stability for 20 seconds with hands on hips and eyes closed. You should inform the child that you will be counting the number of times the child moves out of this position. If the child stumbles out of this position, instruct him/her to open the eyes and return to the start position and continue balancing. You should start timing when the child is set and the eyes are closed.

(c)

Balance testing – types of errors – Parts (a) and (b)

1. Hands lifted off iliac crest
2. Opening eyes
3. Step, stumble, or fall
4. Moving hip into > 30 degrees abduction
5. Lifting forefoot or heel
6. Remaining out of test position > 5 sec

Each of the 20-second trials is scored by counting the errors, or deviations from the proper stance, accumulated by the child. The examiner will begin counting errors only after the child has assumed the proper start position. **The modified BESS is calculated by adding one error point for each error during the two 20-second tests. The maximum total number of errors for any single condition is 10.** If a child commits multiple errors simultaneously, only one error is recorded but the child should quickly return to the testing position, and counting should resume once subject is set. Children who are unable to maintain the testing procedure for a minimum of **five seconds** at the start are assigned the highest possible score, ten, for that testing condition.

OPTION: For further assessment, the same 2 stances can be performed on a surface of medium density foam (e.g., approximately 50cm x 40cm x 6cm).

Tandem Gait^{4,7}

Use a clock (with a second hand) or stopwatch to measure the time taken to complete this task. Instruction for the examiner – **Demonstrate the following to the child:**

*The child is instructed to stand with their feet together behind a starting line (the test is best done with footwear removed). Then, they walk in a forward direction as quickly and as accurately as possible along a 38mm wide (sports tape), 3 meter line with an alternate foot heel-to-toe gait ensuring that they approximate their heel and toe on each step. Once they cross the end of the 3m line, they turn 180 degrees and return to the starting point using the same gait. **A total of 4 trials are done and the best time is retained.** Children fail the test if they step off the line, have a separation between their heel and toe, or if they touch or grab the examiner or an object. In this case, the time is not recorded and the trial repeated, if appropriate.*

Explain to the child that you will time how long it takes them to walk to the end of the line and back.

Coordination examination

Upper limb coordination

Finger-to-nose (FTN) task:

The tester should **demonstrate it to the child**.

"I am going to test your coordination now. Please sit comfortably on the chair with your eyes open and your arm (either right or left) outstretched (shoulder flexed to 90 degrees and elbow and fingers extended). When I give a start signal, I would like you to perform five successive finger to nose repetitions using your index finger to touch the tip of the nose as quickly and as accurately as possible."

Scoring: 5 correct repetitions in < 4 seconds = 1

Note for testers: Children fail the test if they do not touch their nose, do not fully extend their elbow or do not perform five repetitions. **Failure should be scored as 0.**

References & Footnotes

1. This tool has been developed by a group of international experts at the 4th International Consensus meeting on Concussion in Sport held in Zurich, Switzerland in November 2012. The full details of the conference outcomes and the authors of the tool are published in The BJSM Injury Prevention and Health Protection, 2013, Volume 47, Issue 5. The outcome paper will also be simultaneously co-published in other leading biomedical journals with the copyright held by the Concussion in Sport Group, to allow unrestricted distribution, providing no alterations are made.
2. McCrory P et al., Consensus Statement on Concussion in Sport – the 3rd International Conference on Concussion in Sport held in Zurich, November 2008. British Journal of Sports Medicine 2009, 43: 176–89.
3. Maddocks, DL; Dicker, GD; Saling, MM. The assessment of orientation following concussion in athletes. Clinical Journal of Sport Medicine. 1995; 5(1): 32–3.
4. McCrea M. Standardized mental status testing of acute concussion. Clinical Journal of Sport Medicine. 2001; 11: 176–181.
5. Guskiewicz KM. Assessment of postural stability following sport-related concussion. Current Sports Medicine Reports. 2003; 2: 24–30.
6. Schneiders, A.G., Sullivan, S.J., Gray, A., Hammond-Tooke, G. & McCrory, P. Normative values for 16–37 year old subjects for three clinical measures of motor performance used in the assessment of sports concussions. Journal of Science and Medicine in Sport. 2010; 13(2): 196–201.
7. Schneiders, A.G., Sullivan, S.J., Kvarnstrom, J.K., Olsson, M., Yden, T. & Marshall, S.W. The effect of footwear and sports-surface on dynamic neurological screening in sport-related concussion. Journal of Science and Medicine in Sport. 2010; 13(4): 382–386.
8. Ayr, L.K., Yeates, K.O., Taylor, H.G., & Brown, M. Dimensions of post-concussive symptoms in children with mild traumatic brain injuries. Journal of the International Neuropsychological Society. 2009; 15:19–30.

Figure 20.6 (cont.) Child SCAT3™

CHILD ATHLETE INFORMATION

Any child suspected of having a concussion should be removed from play, and then seek medical evaluation. The child must NOT return to play or sport on the same day as the suspected concussion.

Signs to watch for

Problems could arise over the first 24–48 hours. The child should not be left alone and must go to a hospital at once if they develop any of the following:

- New Headache, or Headache gets worse
- Persistent or increasing neck pain
- Becomes drowsy or can't be woken up
- Can not recognise people or places
- Has Nausea or Vomiting
- Behaves unusually, seems confused, or is irritable
- Has any seizures (arms and/or legs jerk uncontrollably)
- Has weakness, numbness or tingling (arms, legs or face)
- Is unsteady walking or standing
- Has slurred speech
- Has difficulty understanding speech or directions

Remember, it is better to be safe.

Always consult your doctor after a suspected concussion.

Return to school

Concussion may impact on the child's cognitive ability to learn at school. This must be considered, and medical clearance is required before the child may return to school. **It is reasonable for a child to miss a day or two of school after concussion, but extended absence is uncommon.** In some children, a graduated return to school program will need to be developed for the child. The child will progress through the return to school program provided that there is no worsening of symptoms. If any particular activity worsens symptoms, the child will abstain from that activity until it no longer causes symptom worsening. Use of computers and internet should follow a similar graduated program, provided that it does not worsen symptoms. This program should include communication between the parents, teachers, and health professionals and will vary from child to child. The return to school program should consider

- Extra time to complete assignments/tests
- Quiet room to complete assignments/tests
- Avoidance of noisy areas such as cafeterias, assembly halls, sporting events, music class, shop class, etc
- Frequent breaks during class, homework, tests
- No more than one exam/day
- Shorter assignments
- Repetition/memory cues
- Use of peer helper/tutor
- Reassurance from teachers that student will be supported through recovery through accommodations, workload reduction, alternate forms of testing
- Later start times, half days, only certain classes

The child is not to return to play or sport until he/she has successfully returned to school/learning, without worsening of symptoms. Medical clearance should be given before return to play.

If there are any doubts, management should be referred to a qualified health practitioner, expert in the management of concussion in children.

Return to sport

There should be no return to play until the child has successfully returned to school/learning, without worsening of symptoms.

Children must not be returned to play the same day of injury.

When returning children to play, they should **medically cleared and then follow a stepwise supervised program**, with stages of progression.

For example:

Rehabilitation stage	Functional exercise at each stage of rehabilitation	Objective of each stage
No activity	Physical and cognitive rest	Recovery
Light aerobic exercise	Walking, swimming or stationary cycling keeping intensity, 70% maximum predicted heart rate. No resistance training	Increase heart rate
Sport-specific exercise	Skating drills in ice hockey, running drills in soccer. No head impact activities	Add movement
Non-contact training drills	Progression to more complex training drills, eg passing drills in football and ice hockey. May start progressive resistance training	Exercise, coordination, and cognitive load
Full contact practice	Following medical clearance participate in normal training activities	Restore confidence and assess functional skills by coaching staff
Return to play	Normal game play	

There should be approximately 24 hours (or longer) for each stage and the child should drop back to the previous asymptomatic level if any post-concussive symptoms recur. Resistance training should only be added in the later stages.

If the child is symptomatic for more than 10 days, then review by a health practitioner, expert in the management of concussion, is recommended.

Medical clearance should be given before return to play.

Notes:

CONCUSSION INJURY ADVICE FOR THE CHILD AND PARENTS / CARERS

(To be given to the **person monitoring** the concussed child)

This child has received an injury to the head. A careful medical examination has been carried out and no sign of any serious complications has been found. It is expected that recovery will be rapid, but the child will need monitoring for the next 24 hours by a responsible adult.

If you notice any change in behavior, vomiting, dizziness, worsening headache, double vision or excessive drowsiness, please call an ambulance to transport the child to hospital immediately.

Other important points:

- Following concussion, the child should rest for at least 24 hours
- The child should avoid any computer, internet or electronic gaming activity if these activities make symptoms worse.
- The child should not be given any medications, including pain killers, unless prescribed by a medical practitioner.
- The child must not return to school until medically cleared.
- The child must not return to sport or play until medically cleared.

Clinic phone number

(d)

Patient's name

Date/time of injury

Date/time of medical review

Treating physician

Contact details or stamp

Figure 20.6 (cont.) Child SCAT3™

School attendance and activities should be considered and modified as appropriate to avoid provocation of symptoms.

Due to the different physiological response and longer recovery after concussion, and specific risks (e.g. diffuse cerebral swelling) related to head impact during childhood and adolescence, a more conservative RTP

approach is recommended.²² It is appropriate to extend the amount of time of asymptomatic rest and/or the length of the graded exertion in children and adolescents. In the management paradigm of children post concussion, return to school should be achieved prior to consideration of return to sport.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e